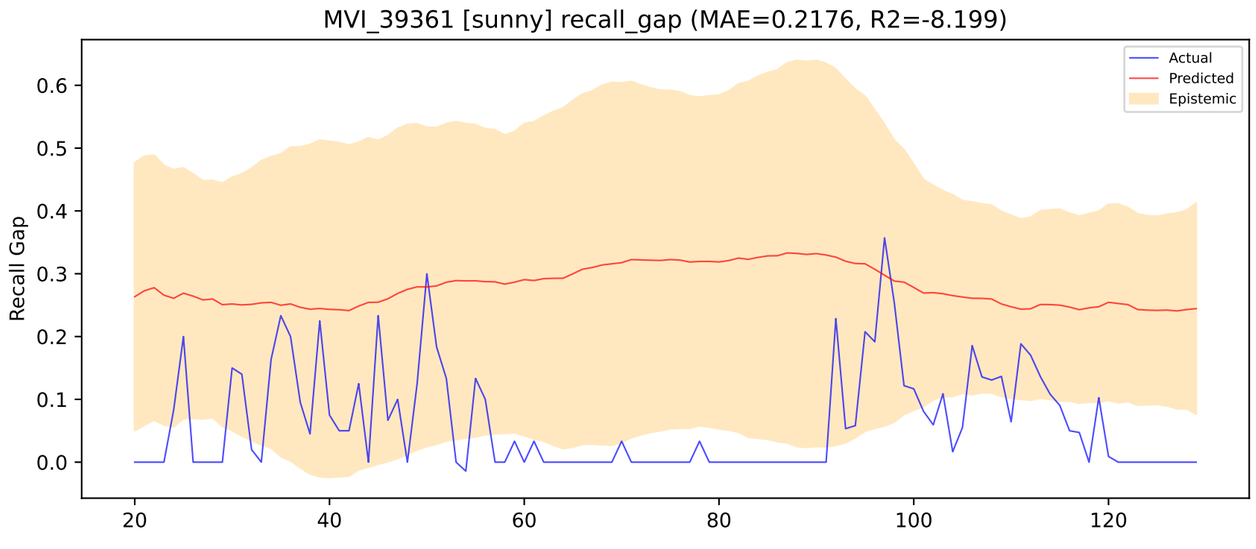
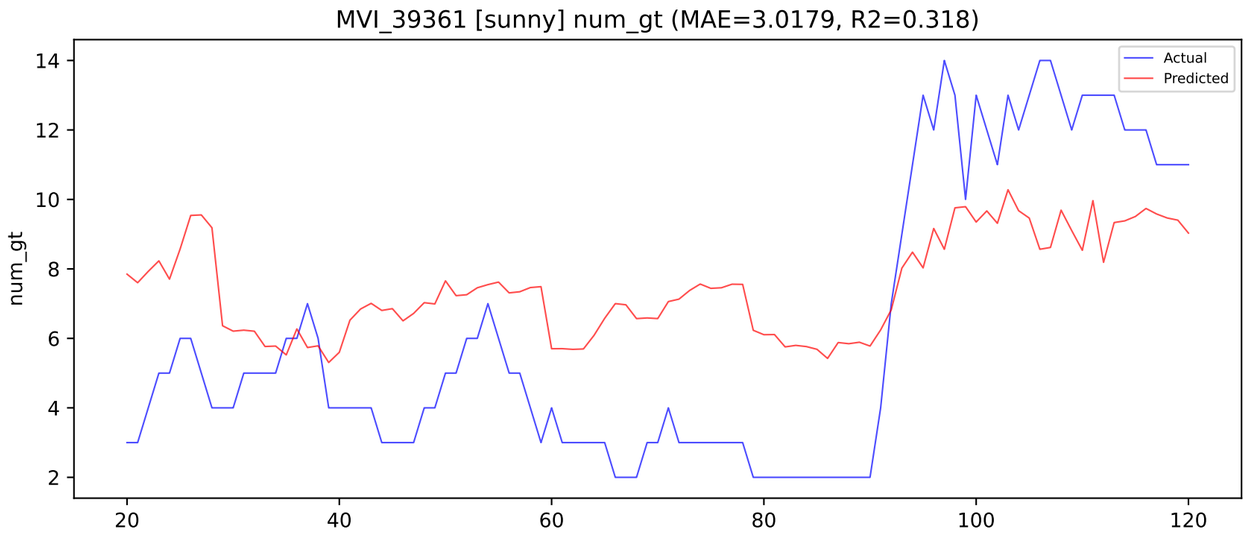
Page 1: DETRAC Validation — Top 2 Sequences per Weather

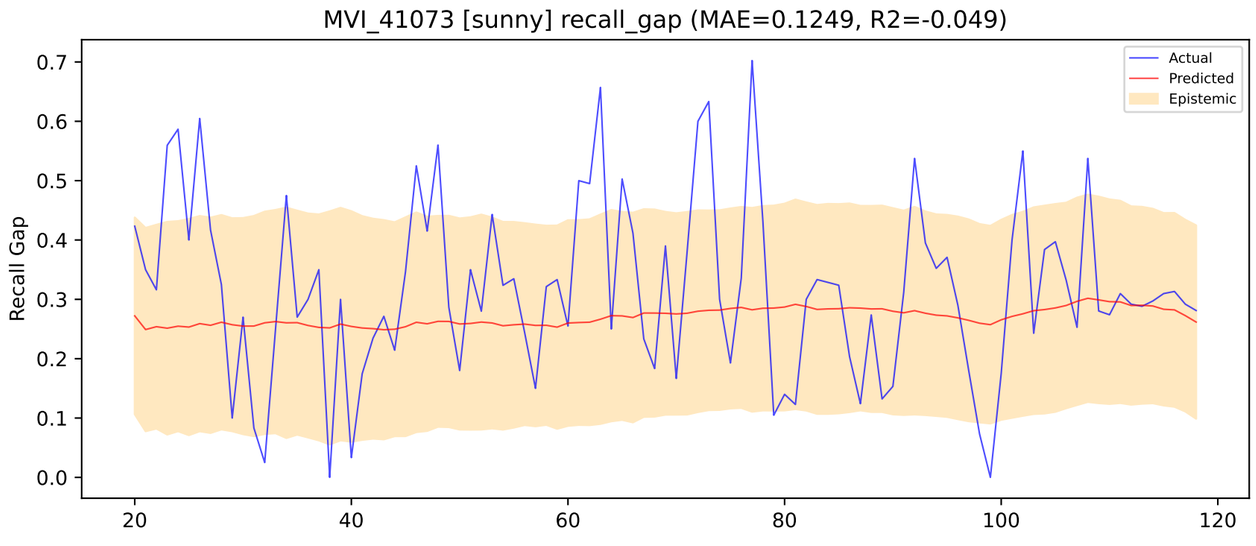
MVI_39361 [sunny] recall_gap (MAE=0.2176, R2=-8.199)



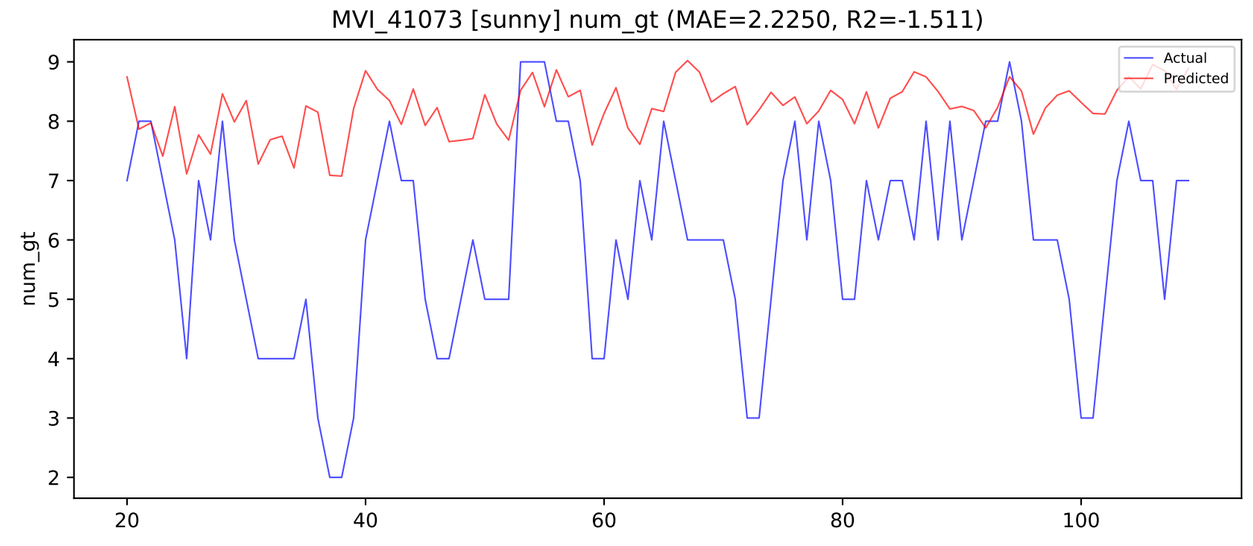
MVI_39361 [sunny] num_gt (MAE=3.0179, R2=0.318)



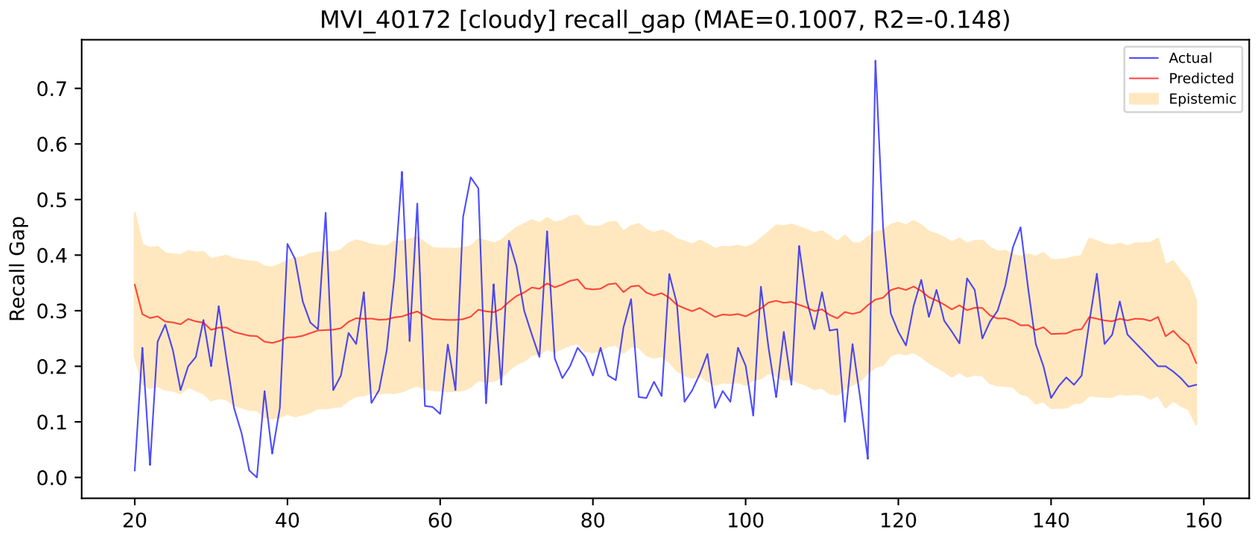
MVI_41073 [sunny] recall_gap (MAE=0.1249, R2=-0.049)



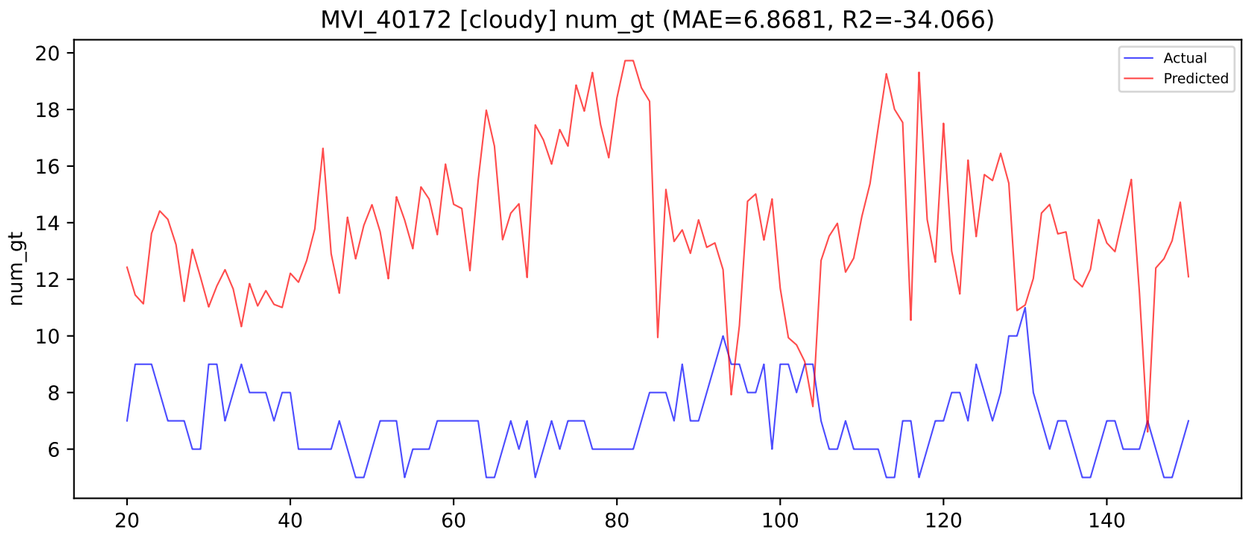
MVI_41073 [sunny] num_gt (MAE=2.2250, R2=-1.511)



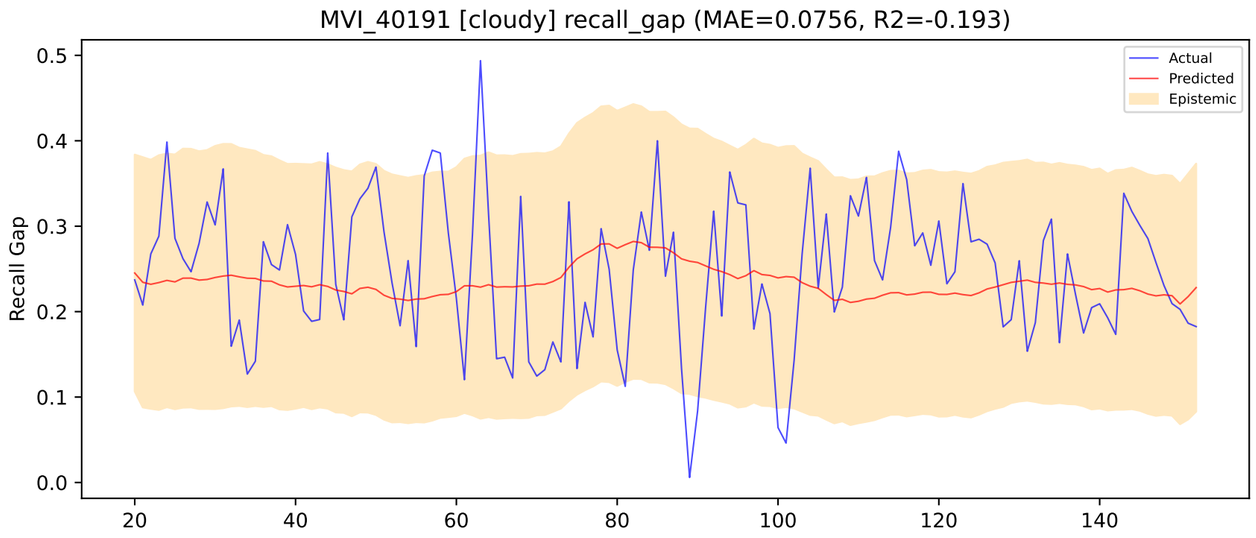
MVI_40172 [cloudy] recall_gap (MAE=0.1007, R2=-0.148)



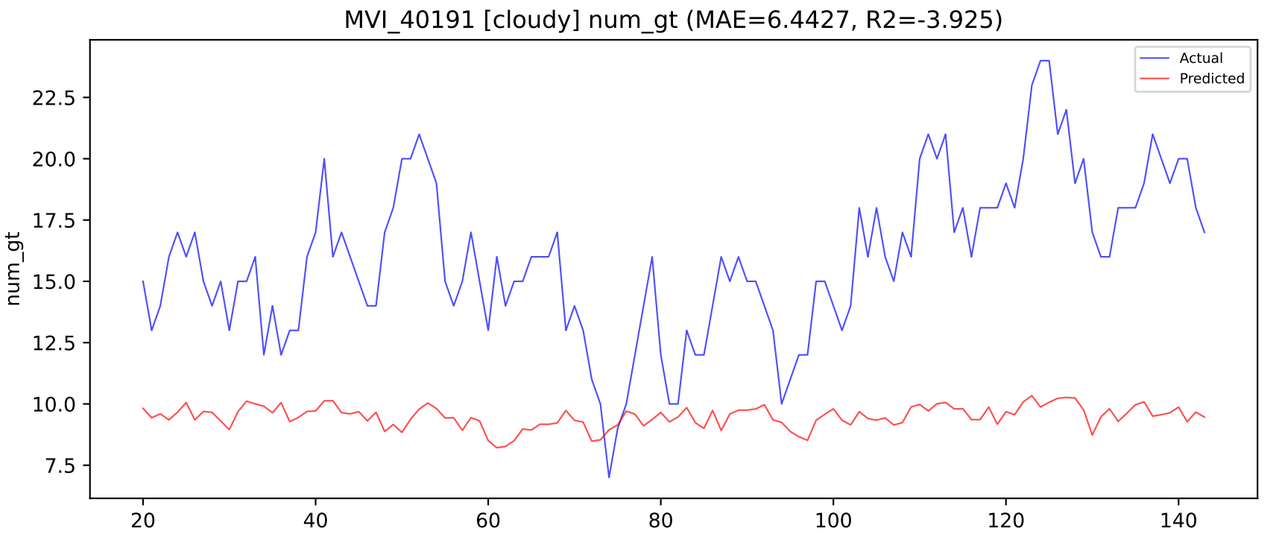
MVI_40172 [cloudy] num_gt (MAE=6.8681, R2=-34.066)



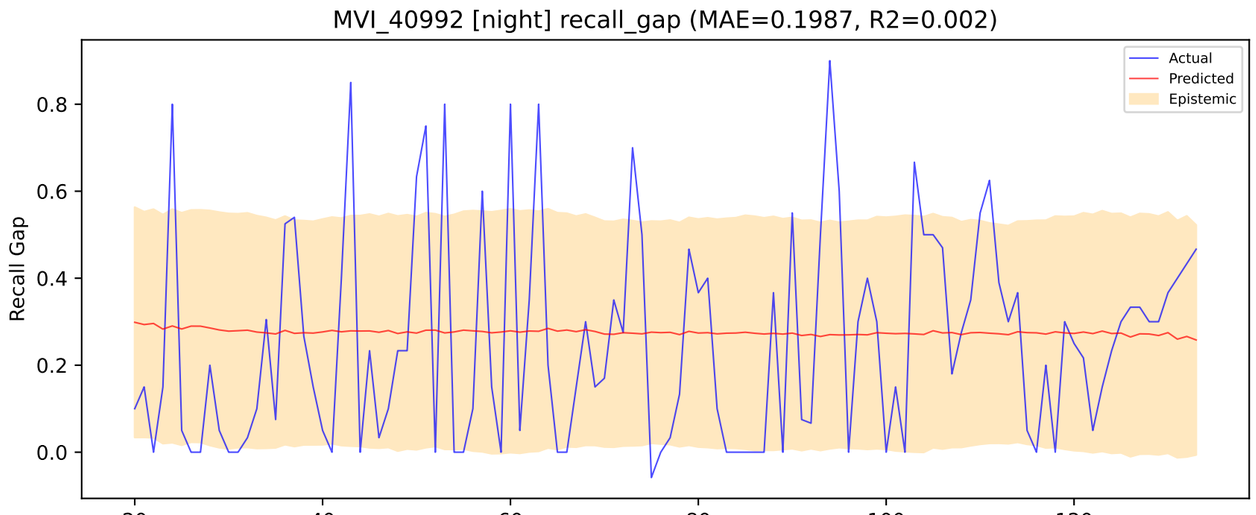
MVI_40191 [cloudy] recall_gap (MAE=0.0756, R2=-0.193)



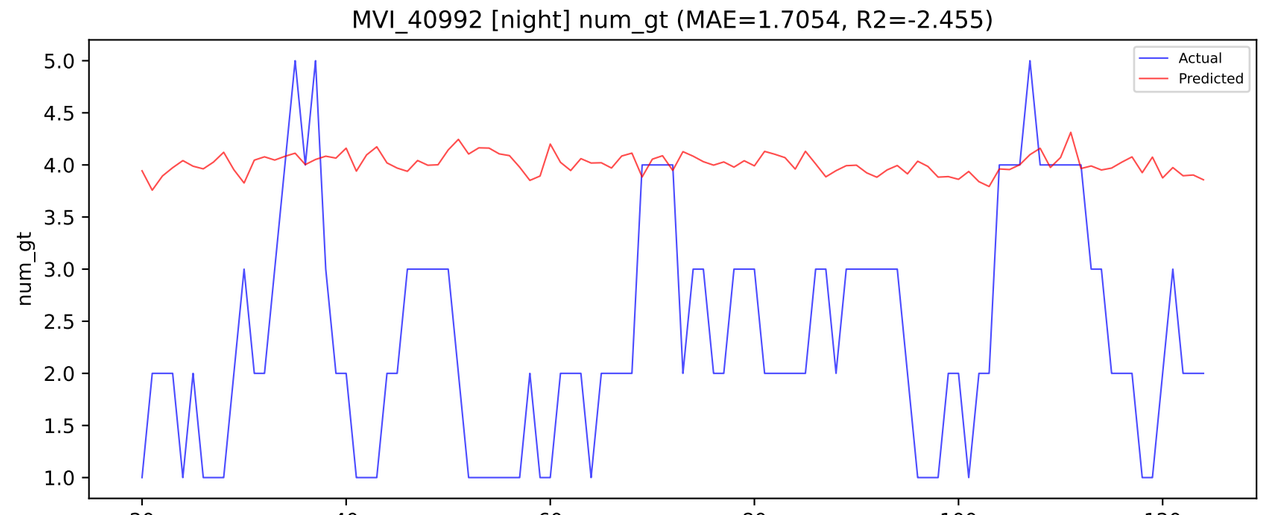
MVI_40191 [cloudy] num_gt (MAE=6.4427, R2=-3.925)



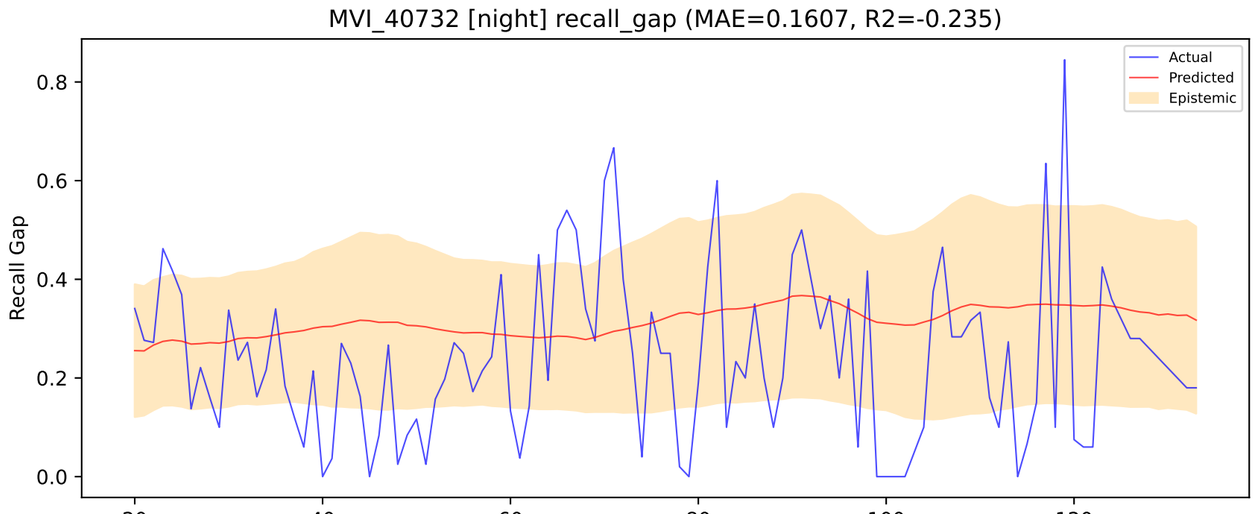
MVI_40992 [night] recall_gap (MAE=0.1987, R2=0.002)



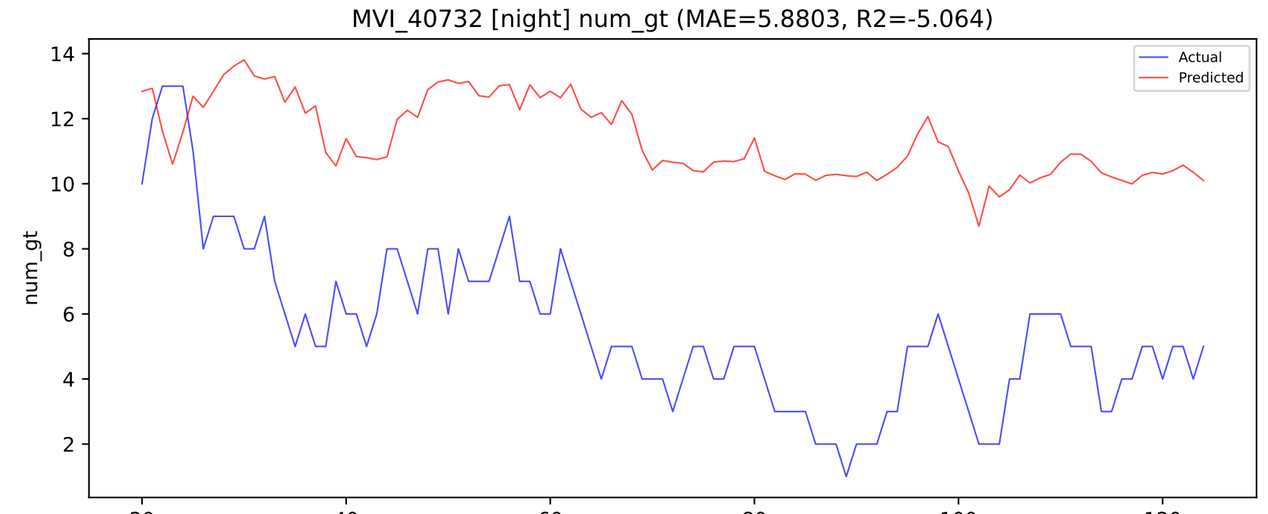
MVI_40992 [night] num_gt (MAE=1.7054, R2=-2.455)



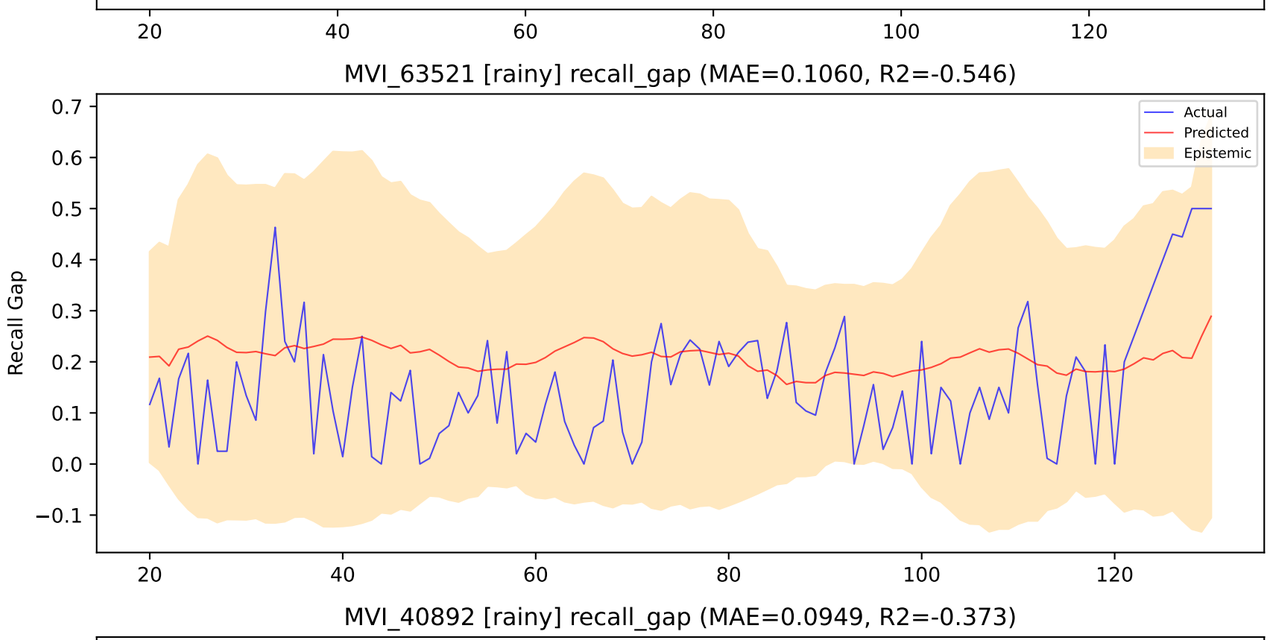
MVI_40732 [night] recall_gap (MAE=0.1607, R2=-0.235)



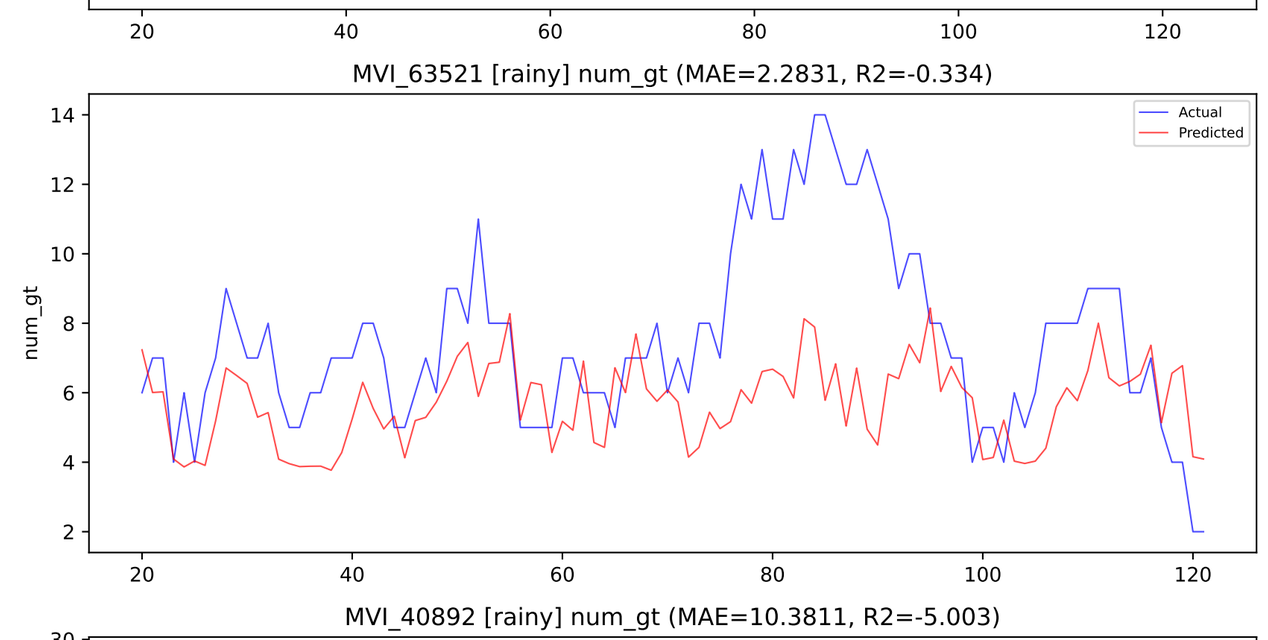
MVI_40732 [night] num_gt (MAE=5.8803, R2=-5.064)



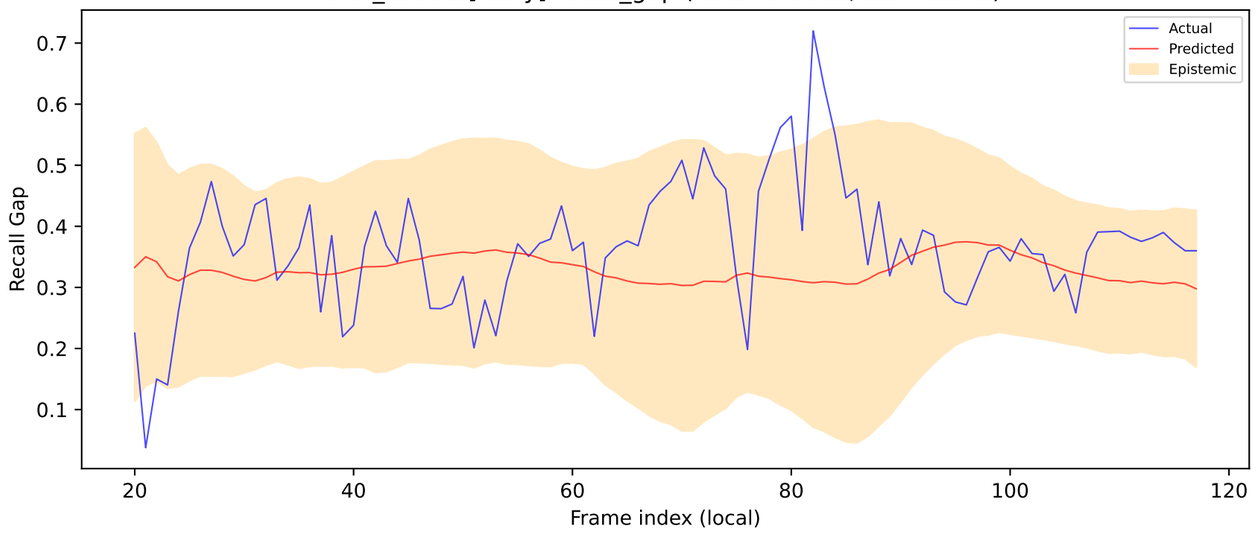
MVI_63521 [rainy] recall_gap (MAE=0.1060, R2=-0.546)



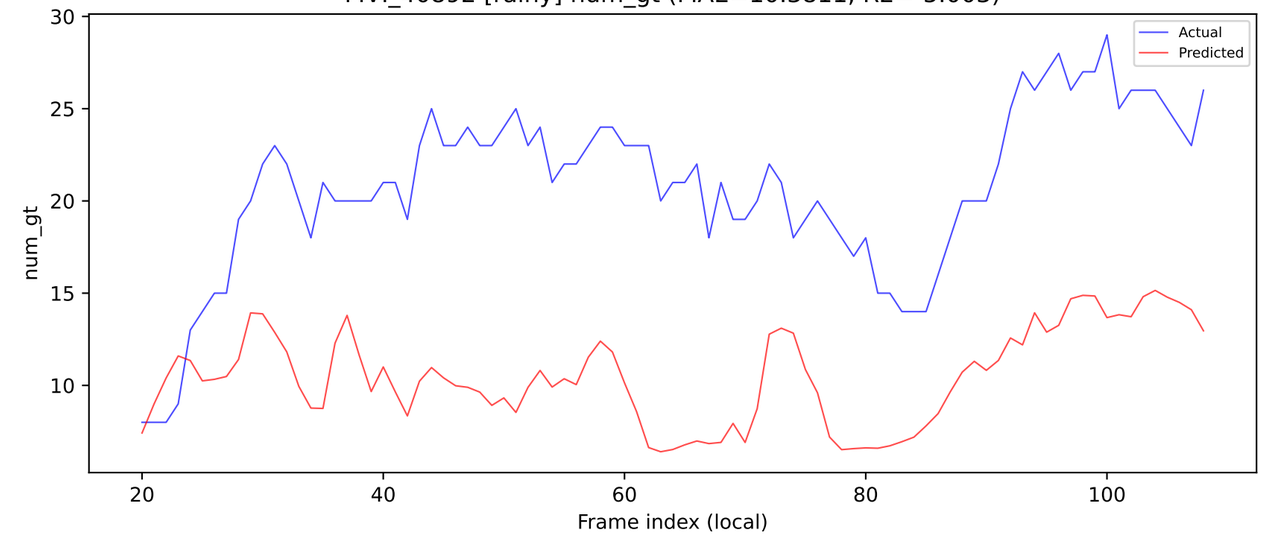
MVI_63521 [rainy] num_gt (MAE=2.2831, R2=-0.334)



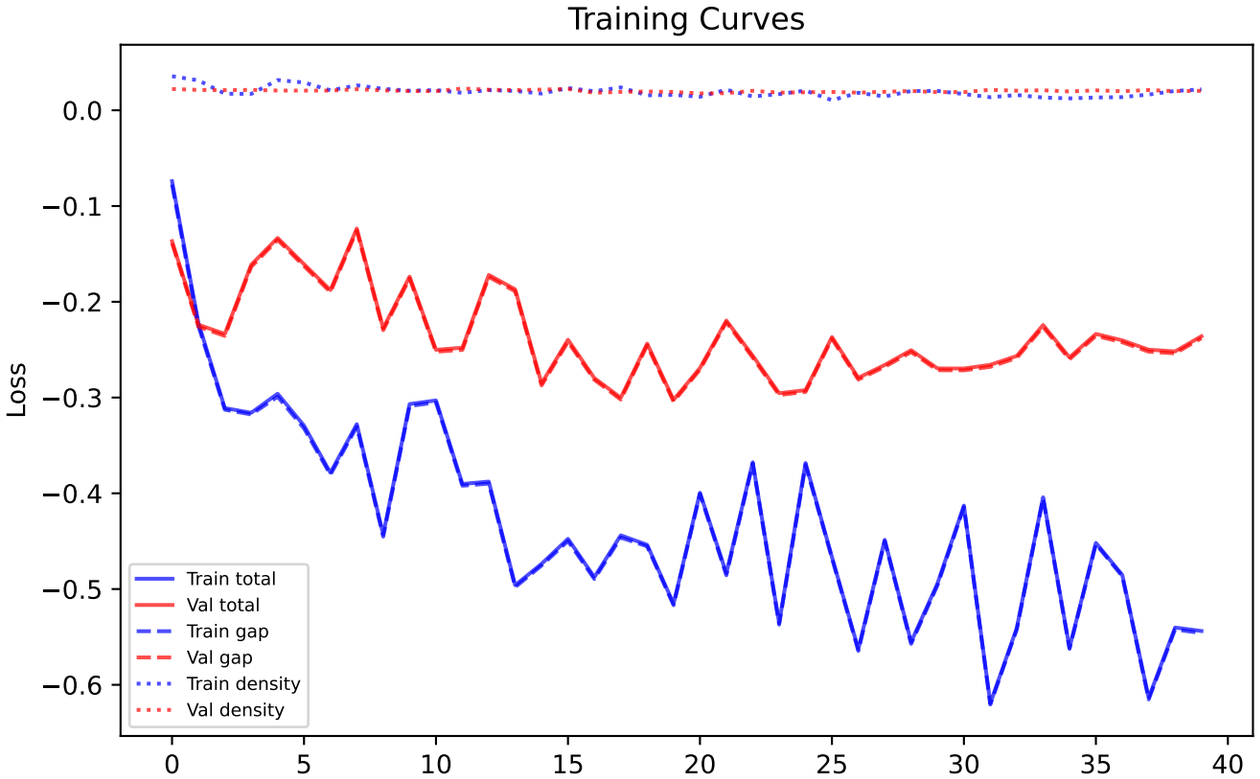
MVI_40892 [rainy] recall_gap (MAE=0.0949, R2=-0.373)

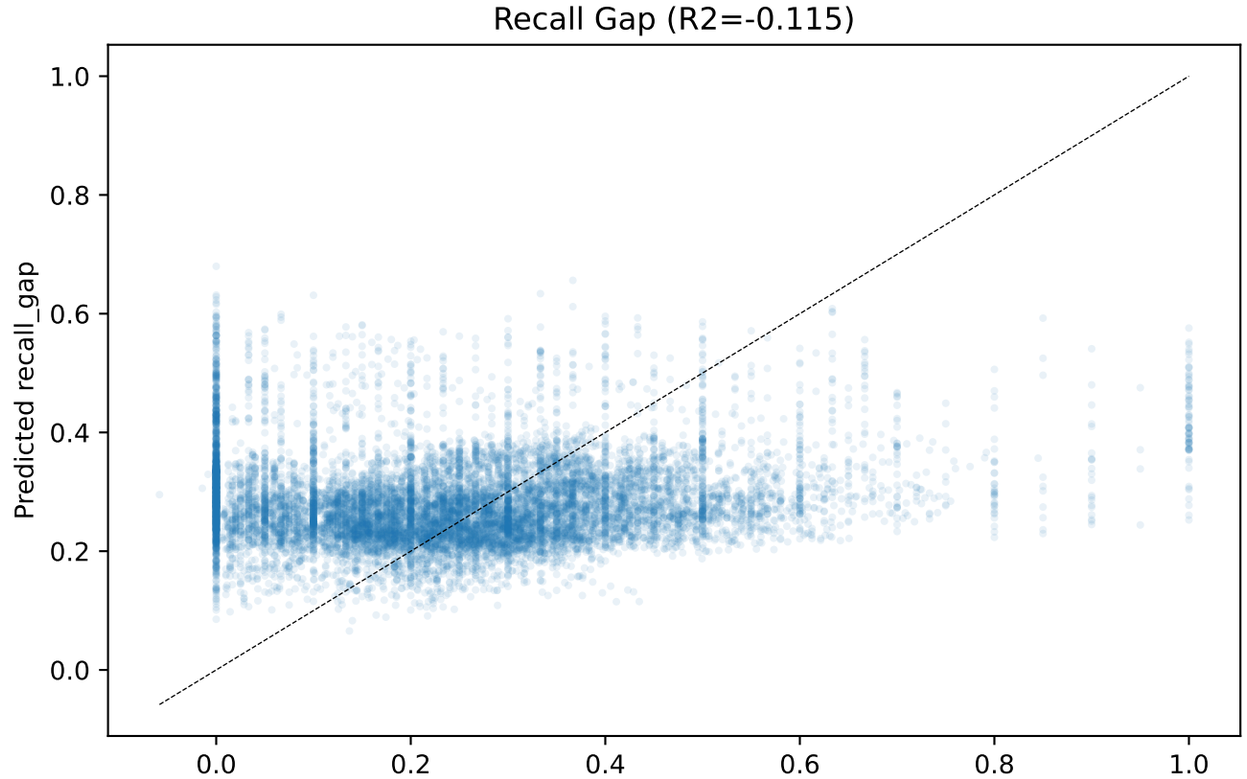


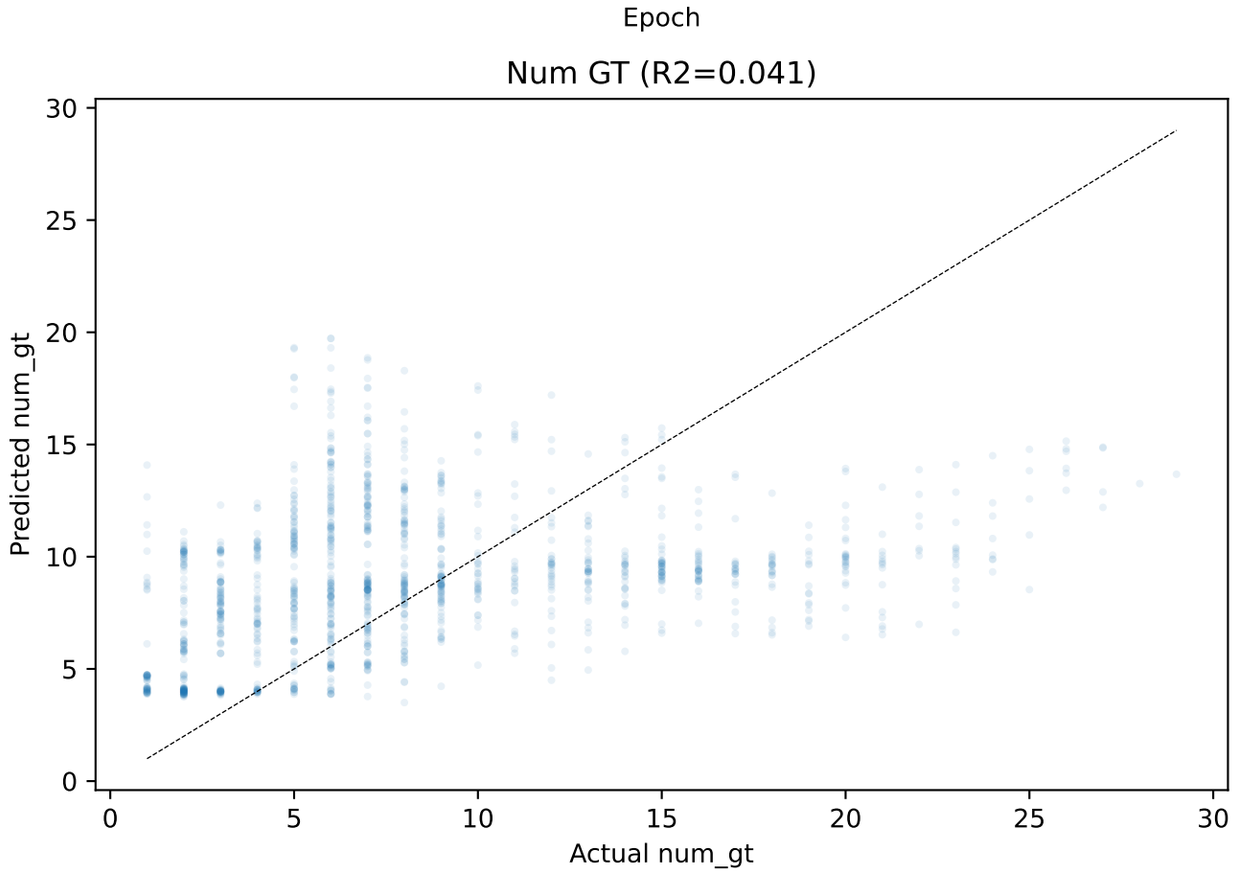
MVI_40892 [rainy] num_gt (MAE=10.3811, R2=-5.003)

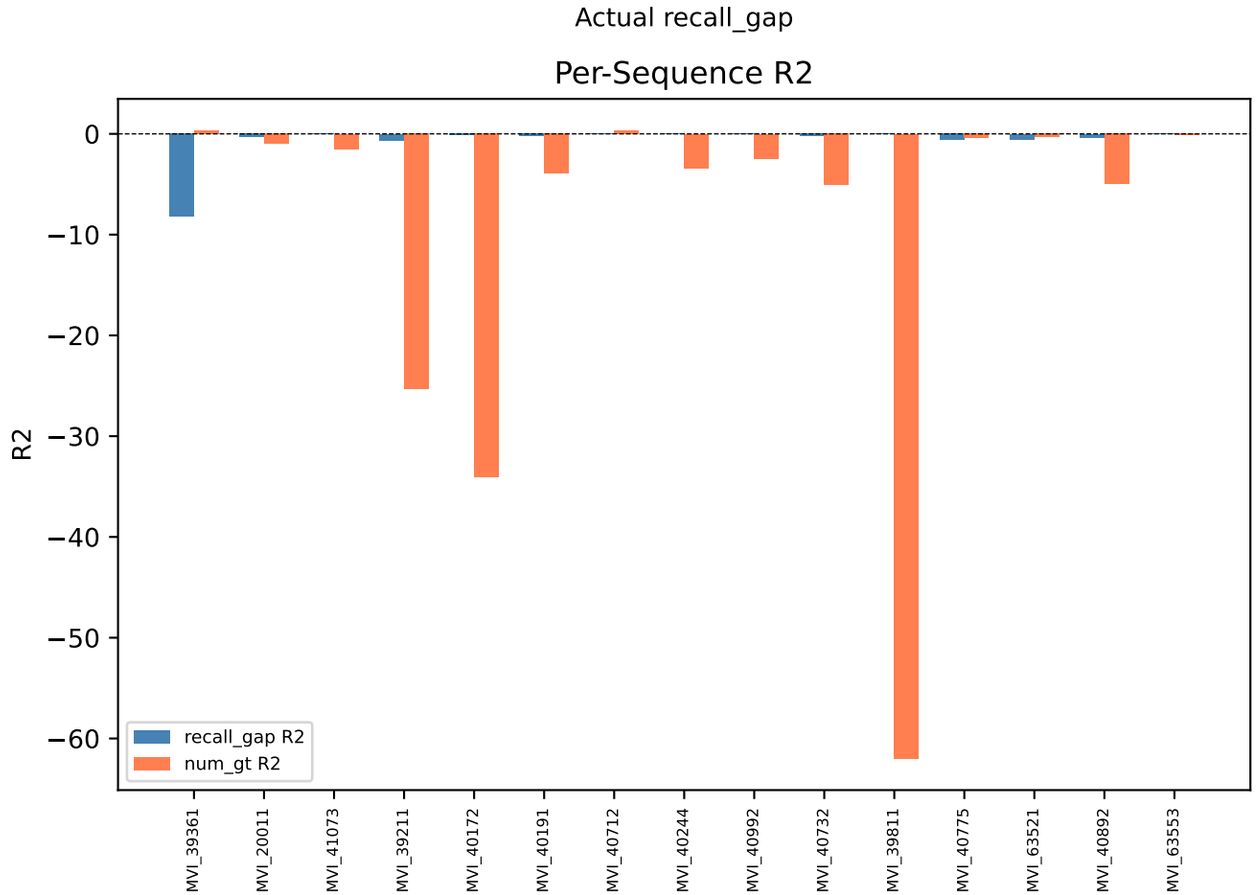


Page 2: Diagnostics









Page 3: Metrics & Comparison

DETRAC LSTM Validation Metrics (all 15 sequences)

Sequence	Weather	Gap MAE	Gap R2	Gap rho	GT MAE	GT R2	GT rho
MVI_39361	sunny	0.2176	-8.1989	-0.4685	3.0179	0.3180	0.6978
MVI_20011	sunny	0.1149	-0.3199	-0.1148	3.0921	-0.9820	0.0335
MVI_41073	sunny	0.1249	-0.0494	0.1444	2.2250	-1.5112	0.4310
MVI_39211	sunny	0.3160	-0.6416	0.0060	6.4492	-25.3358	-0.1899
MVI_40172	cloudy	0.1007	-0.1485	0.1856	6.8681	-34.0658	-0.4450
MVI_40191	cloudy	0.0756	-0.1930	-0.0751	6.4427	-3.9249	0.4228
MVI_40712	cloudy	0.1638	0.0369	0.1234	2.7653	0.3521	0.8199
MVI_40244	cloudy	0.1041	-0.0125	0.0032	4.4037	-3.4062	0.1253
MVI_40992	night	0.1987	0.0016	0.0483	1.7054	-2.4552	-0.0228
MVI_40732	night	0.1607	-0.2350	-0.0614	5.8803	-5.0645	0.7680
MVI_39811	night	0.3105	-0.0156	-0.1133	3.4083	-62.0036	-0.0556
MVI_40775	night	0.0977	-0.5956	0.0301	5.2419	-0.4167	0.8179
MVI_63521	rainy	0.1060	-0.5464	0.0224	2.2831	-0.3344	0.4640
MVI_40892	rainy	0.0949	-0.3733	-0.2485	10.3811	-5.0035	0.5629
MVI_63553	rainy	0.0851	-0.0083	0.1009	1.5756	-0.1516	-0.1370
OVERALL		0.1444	-0.1151	0.0840	4.5354	0.0406	0.3889
BASELINE	(mean)	0.1375	0.0000	nan	4.5528	0.0000	nan

Per-Weather Gap R² + UAVDT Comparison

Weather/Source	Gap R2	Gap MAE	n (windows)
sunny	-0.4130	0.20410	3000
cloudy	-0.0340	0.11120	4400
night	-0.0517	0.17689	2820
rainy	0.2605	0.09665	2600
OVERALL	-0.1151	0.14444	12820
UAVDT js10	-0.2700	—	—